

Project Overview

This project will manage intern data and batches for a company or organization. It will include features such as intern registration, ID card generation, batch management, and displaying intern and batch details. Interns will be assigned to batches, and each intern will have a unique ID based on their date of joining.

1. Functional Requirements

1.1 Intern Management

- **Add Intern:**
 - Interns can be added by providing the following details:
 - Name (String)
 - Email (String)
 - Mobile Number (String)
 - ID Card Type (Select from “Free” or “Premium”)
 - Date of Joining (Date)
 - Batch Assignment (Dropdown of available batches)
 - **Generate Intern ID:**
 - Intern ID should be auto-generated based on the following format:
 - **For Free ID Card:** TDA{DateOfJoining in YYYYMMDD}-{ThreeDigitNumber}
 - **For Premium ID Card:** EMP{DateOfJoining in YYYYMMDD}-{ThreeDigitNumber}
 - Example: EMP20241129-001 or TDA20241129-001
 - **Intern List:**
 - Display a list of all interns with the following columns:
 - Intern ID
 - Name
 - Email
 - Mobile Number
 - Batch
 - ID Card Type
 - Date of Joining
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1.2 Batch Management

- **Create Batch:**
 - Batches can be created by providing the following details:
 - Start Date (Date)
 - End Date (Date) – Automatically calculated as 6 months after the start date.

- **Assign Interns to Batch:**
 - Interns should be assigned to a batch during registration.
 - **Batch List:**
 - Display a list of all batches with the following details:
 - Batch Start Date
 - Batch End Date
 - Total Number of Interns in the Batch
 - Intern Details (Name, ID Card Type, etc.)
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1.3 Additional Features

- **Update Intern Details:**
 - Ability to update intern details (e.g., name, email, mobile number) but not the intern ID once generated.
 - **Delete Intern:**
 - Ability to delete intern records.
 - **Batch Overview:**
 - Overview of batch performance, with a list of all interns assigned and their details.
 - **Search and Filter:**
 - Ability to search and filter intern data based on:
 - Name
 - Batch
 - ID Card Type
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2. Non-Functional Requirements

2.1 Backend (Java - Spring Boot)

- **API Development:**
 - The backend should expose RESTful APIs to handle operations such as:
 - Adding an intern
 - Generating an intern ID
 - Creating batches
 - Assigning interns to batches
 - Viewing intern and batch details
- **Database:**
 - Use a relational database (e.g., MySQL, PostgreSQL) to store intern and batch information.

- Intern and Batch data should be stored in separate tables and linked by a foreign key relationship.
- **ID Generation Logic:**
 - Implement a unique ID generation logic that ensures each intern has a distinct ID based on their joining date and ID card type.

2.2 Frontend (Angular)

- **UI Components:**
 - **Intern Registration Form:** A form to capture intern details and assign them to a batch.
 - **Batch Creation Form:** A form to create a new batch, with automatic calculation of end date (6 months after start).
 - **Intern List View:** A table to view all interns with filtering options (e.g., by batch or ID card type).
 - **Batch List View:** A table to view all batches with intern count and batch details.
 - **Validation:**
 - Validate intern data (e.g., ensure email and mobile number are correctly formatted).
 - Ensure no intern is added without being assigned to a batch.
 - **Responsive Design:**
 - The frontend must be responsive, accessible, and easy to use on both desktop and mobile devices.
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3. System Design

- **Batch End Date Calculation:**
 - The batch end date should be automatically set to 6 months after the batch start date.
 - **ID Card Generation:**
 - The ID generation logic should ensure the correct format (e.g., EMP20241129-001 or TDA20241129-001).
 - The numbering should be sequential per batch (e.g., 001, 002, 003).
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4. Technology Stack

4.1 Backend (Java):

- **Framework:** Spring Boot
- **Database:** MySQL or PostgreSQL
- **API:** RESTful APIs
- **ID Generation:** Java UUID or custom ID generation logic

4.2 Frontend (Angular):

- **Framework:** Angular (latest version)
- **UI Framework:** Angular Material (for UI components like tables, forms)
- **HTTP Client:** Axios or Angular HTTP client for API calls

- **Form Validation:** Angular reactive forms or template-driven forms
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5. Deliverables

1. Backend:

- RESTful APIs for managing interns and batches.
- Database schema for interns and batches.
- ID generation and validation logic.

2. Frontend:

- Angular application with pages to manage interns and batches.
 - UI for adding, updating, and deleting intern and batch records.
 - Display intern and batch details in tabular format.
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6. Timeline and Milestones

• Phase 1 (Week 1-2):

- Set up backend services and API endpoints for intern and batch management.
- Set up Angular frontend project and create components for intern and batch management.

• Phase 2 (Week 3-4):

- Implement intern and batch CRUD operations.
- Implement ID card generation logic.
- Integrate backend API with Angular frontend.

• Phase 3 (Week 5-6):

- Testing of APIs and frontend UI.
 - Validation and filtering for intern data.
 - Deployment preparation.
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7. Optional Future Enhancements

- **Intern Performance Tracking:** Track intern performance through an evaluation system.
- **Notifications:** Email or SMS notifications when interns are assigned to batches or batches are created.
- **Role-based Access:** Implement user roles (Admin, Manager) for different levels of access to the system.